Vivek Kumar

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EDUCATION

University of Minnesota Twin Cities Minneapolis, MN
Master of Science in Robotics *Sep 2024 – Present*
Relevant Coursework: *Machine learning, Computer Vision, Deep Learning, Intelligent Robotic Systems*

National Institute of Technology Rourkela Odisha, India
Bachelor of Technology in Mechanical Engineering *2019 – 2023*

SKILLS

Programming & Frameworks: Python, C++, ROS 2, PyTorch, OpenCV, TensorRT, ONNX, Hugging Face
AI: Vision-Language Models (VLMs), Vision-Language-Action (VLAs), Diffusion Models, Imitation Learning, RL
Robotics & Hardware: SLAM, Path Planning, Nav2, Mapping, Sensor Fusion, Camera, LiDAR
Simulation & Tools: Isaac Sim, Gazebo, Rviz, MoveIt, Docker, SolidWorks, Linux

EXPERIENCE

NextGX Lab, University of Minnesota | Graduate Research Assistant Minneapolis, MN
Autonomous Driving, Vision-Language-Action Models, RL, AMR, VLMs *Jun 2025 – Present*

- Developing a sign-guided **Vision-Language-Action** framework for long-horizon **mapless navigation** that leverages semantic scene understanding and language reasoning to navigate toward user-specified destinations.
- Developed REACT2, a **reasoning model** using Qwen3-VL for multi-robot supervision, fine-tuned it on custom dataset containing **6000+** preference pairs covering long-tail autonomous mobile robots scenarios.
- Designed a **reinforcement learning**-based alert system uisng **PPO** for an MnDOT-funded intersection-safety project, fusing vehicle-to-vehicle interaction data to identify and flag high-risk traffic conflicts in real time.
- Built a VLM model for proactive escalation in autonomous driving using Qwen3-VL and **NVIDIA Cosmos** as teacher model, fine-tuned on a **nuScenes** dataset with edge-case based sampling via **LoRA** and **DPO**, achieving **92.1%** accuracy.
- Developed an **ADAS** stack using YOLOv8m for **detection**, ByteTrack for **tracking**, UFLD-v2 for **lane detection** and **BEV** uisng IPM achieving **15 FPS** real-time inference via TensorRT/ONNX acceleration.

GLOVEX Lab, University of Minnesota | Graduate Researcher Minneapolis, MN
Inverse Problems, Diffusion and Flow-matching models *April 2025 – Present*

- Conducting research on inverse problems using generative priors like **diffusion** and **flow-matching** models.
- Built ZS-PDP, a novel zero-shot blind inpainting framework, outperforming both **SOTA** supervised and zero-shot methods.

PUBLICATIONS

- **Vivek Kumar**, et al. *VLM-Driven Proactive Escalation for Tele-operations in Autonomous Vehicles.* **CVPR Precognition Workshop 2026.**
- **Vivek Kumar**, et al. *Robots That Know When to Ask: VLM-Driven Reasoning for Human-on-Demand Multi-Robot Teleoperation.* (Under Review) **Demo**
- W Zhang, **Vivek Kumar**, et al. *Zero-Shot Blind Inpainting with Pretrained Diffusion Priors.* **NeurIPS 2026** (Under Review)

PROJECTS

Full-Stack Autonomous Navigation System for AgileX Hunter 2.0 | Demo *ROS 2, C++, Python*

- Developed a production-ready modular autonomy stack on real hardware using ROS 2 by integrating **SLAM**, LiDAR **localization**, D*+ **path planning**, and PID-based control for autonomous indoor navigation in pre-mapped environments.
- Built the collision avoidance pipeline using occupancy grid maps and **octomap** for navigation in dynamic environments.

Multimodal Diffusion Policy for Autonomous Driving *Python, PyTorch*

- Trained a lightweight Vision-Language-Action **(VLA) diffusion policy** on a custom driving dataset, utilizing a CNN-Transformer architecture to predict autonomous trajectories from natural language instructions.

Robot Manipulation via Action Chunking Transformers in ManiSkill *Python, PyTorch, ManiSkill*

- Built an end-to-end **imitation learning** pipeline using Action Chunking Transformer (ACT) architecture to train robot arm manipulation policy on RGBD observations, achieving **95%+** success rate on PickCube task with 30K training iterations.

On-Demand Mobile EV Charging Robot (WEZZON) *ROS 2, Agile-X*

- Built an autonomous mobile EV charging robot to mitigate fixed infrastructure barriers and validated it through real-world parking lot testing using a battery-integrated AgileX Hunter platform.

Predictive Video Generation for Low-Latency Teleoperation *Python, PyTorch*

- Implemented and adapted a **real-time** video prediction model for autonomous driving by training it on the **BDD100K** to mitigate **100ms+** network latency through future-frame prediction.